

Answer Key — Non-contact Forces

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (c) **gravitational force** — gravity acts without contact — non-contact.
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- Q2** (b) **repel** — like poles repel.
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- Q3** (a) **always attractive** — gravity only pulls — it is always attractive.
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- Q4** (b) **electrostatic force** — the rubbed scale is charged and exerts an electrostatic force.
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- Q5** (a) **Both true, R explains A** — the balloons carry like charges, so they repel — exactly as R states.
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- Q6** **negative** — charges are positive and negative.
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- Q7** **gravity** — this pull is gravity.
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Short answer

- Q8** A force felt even without contact between objects; magnetic, electrostatic and gravitational forces.
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- Q9** Like charges/poles repel; unlike charges/poles attract.
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- Q10** Gravity is always attractive (only pulls), while magnetic and electrostatic forces can attract or repel; all three are non-contact.
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Long / application

- Q11** Rubbing gives the balloon and the cloth opposite kinds of charge, so they attract (unlike charges attract). Two balloons rubbed the same way carry like charges, so they repel (like charges repel).
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- Q12** (i) friction — contact (needs surfaces touching); (ii) magnetic — non-contact (acts at a distance); (iii) gravity — non-contact (Earth pulls without touching); (iv) muscular — contact (acts through body parts in contact).
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